of the way they are arranged. We can follow any order.

(i) First, the charge $+q$ is brought to the corner P. This requires no work since no charge is already present, $W_P = 0$

(ii) Work required to bring the charge $-q$ to the corner Q = $(-q) \times$ potential at a point Q due to $+q$ located at a point P.

$$W_Q = -q \times \frac{1}{4\pi\epsilon_\circ}\frac{q}{a} = -\frac{1}{4\pi\epsilon_\circ}\frac{q^2}{a}$$

(iii) Work required to bring the charge $+q$ to the corner R = $q \times$ potential at the point R due to charges at the point P and Q.

$$W_R = q \times \frac{1}{4\pi\epsilon_\circ}\left(-\frac{q}{a} + \frac{q}{\sqrt{2}a}\right)$$
$$= \frac{1}{4\pi\epsilon_\circ}\frac{q^2}{a}\left(-1 + \frac{1}{\sqrt{2}}\right)$$

(iv) Work required to bring the fourth charge $-q$ at the position S = $q \times$ potential at the point S due the all the three charges at the point P, Q and R

$$W_S = -q \times \frac{1}{4\pi\epsilon_\circ}\left(\frac{q}{a} + \frac{q}{a} - \frac{q}{\sqrt{2}a}\right)$$
$$W_S = -\frac{1}{4\pi\epsilon_\circ}\frac{q^2}{a}\left(2 - \frac{1}{\sqrt{2}}\right)$$

(b) Work required to bring the charge q' to the centre of the square = $q' \times$ potential at the centre point O due to all the four charges in the four corners

The potential created by the two $+q$ charges are canceled by the potential created by the $-q$ charges which are located in the opposite corners. Therefore the net electric potential at the centre O due to all the charges in the corners is zero.

Hence no work is required to bring any charge to the point O. Physically this implies that if any charge q' when brought close to O, then it moves to the point O without any external force.

1.5.7 Electrostatic potential energy of a dipole in a uniform electric field

Consider a dipole placed in the uniform electric field $\vec{E}$ as shown in the Figure 1.29. A dipole experiences a torque when kept in a uniform electric field $\vec{E}$. This torque rotates the dipole to align it with the direction of the electric field. To rotate the dipole (at constant angular velocity) from its initial angle θ' to another angle θ against the torque exerted by the electric field, an equal and opposite external torque must be applied on the dipole.

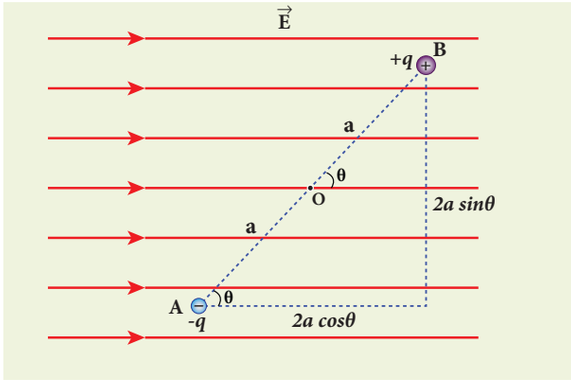


Figure 1.29 The dipole in a uniform electric field

The work done by the external torque to rotate the dipole from angle θ' to θ at constant angular velocity is

$$W = \int_{\theta'}^{\theta} \tau_{ext}\, d\theta \qquad (1.49)$$

Since $\vec{\tau}_{ext}$ is equal and opposite to $\vec{\tau}_E = \vec{p} \times \vec{E}$, we have

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